

Problem Set 1

Due on Canvas at 12:00 noon Friday September 13

Instructions: **No partial credit will be awarded on quantitative questions.** All answers must be entered in the Problem Set 1 Quiz on Canvas. Note that it is **not possible to edit your responses** on Canvas after submitting. Best practice is to **complete the problem set on paper** and then transfer solutions to Canvas. You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. Note that after submitting on canvas, you may see a score, but that does not include any points for any qualitative questions and is only a preliminary score. Grades will be finalized a few days after the due date.

Material covered: This problem set covers the first six sessions of the course (August 27-Sept 12).

Problem 1. Boycotts

Note: This problem has been a tradition at Yale SOM for many years. It is a rite of passage for SOM economics students. To answer this problem requires no mathematics beyond arithmetic. Read the instructions carefully, and then feel free to talk about this problem with your fellow students. We advise such discussions to prevent you from wasting time. In the end you must see the answer yourself in your own “aha” moment and of course, the written answer must be your own.

Setup A frequently used instrument of economic coercion is the boycott. For example, a union involved in a serious and protracted conflict with some employer or group of employers may urge all of its sympathizers to stop buying the products of that employer or group, in the hope that the resulting economic losses will provide an incentive for the employers to make concessions. However, experience shows that the coercive power of a boycott is often small.

In this exercise, you are to use supply and demand analysis as a tool to understand the effects of boycotts. For the sake of concreteness and precision, most of the questions are posed in quantitative terms. However, the goal of the exercise is to develop your understanding of the logic of the market processes involved. Accordingly, if you have difficulty, don't try to overwhelm the problem with calculation -- rethink the economics.

Throughout the exercise, the following assumptions are to be maintained. First, the market in question is one to which simple supply and demand analysis is applicable: there are numerous buyers and sellers in the market; the good is homogeneous; buyers and sellers seek only to trade on the most advantageous possible terms and, aside from the complications introduced by the boycott, do not care with whom they transact. Second, the identities of buyers and sellers are fixed: in particular, no reselling of the goods takes place. Third, the set of buyers divides sharply into observers and nonobservers of the boycott, while the sellers separate equally sharply into those that are targets of the boycott and those that are not. Boycott observers will not, under any circumstances, buy from a target seller, while non-observers do not care from whom they buy. For simplicity, it is assumed that boycott observers account for the same fraction of demand at every price. Similarly, target sellers account for a certain fraction of the supply at every price.

Assume initially that in the pre-boycott situation, the market demand curve is described by:

$$Q_D = 5000 - 10P$$

where Q_D = quantity demanded and P = the price. The market supply curve has equation (with Q_S = quantity supplied):

$$Q_S = 50(P - 5)$$

except that, at prices of 5 or less, quantity supplied is zero.

Warmup problem (i): Determine the market equilibrium in the pre-boycott situation.

Warmup problem (ii): Suppose that buyers representing 70% of the original demand, observe the boycott. If all non-observers buy from target sellers, what is the demand curve facing target sellers?

Questions

Consider the three situations characterized by the following table

<u>Situation</u>	<u>Observers as Fraction of Original Demand</u>	<u>Target as Fraction of Original Supply</u>
A	$3/4$	$3/4$
B	$1/4$	$1/4$
C	$1/2$	$1/4$

For each of the three situations, determine the market equilibrium that would arise under boycott conditions. Give the quantities transacted by observing and non-observing buyers, and by target and non-target sellers.

Drawing on your results, state the general principle which relates the effectiveness of a boycott to the observance fraction and the target fraction. "Effectiveness," in this context, means lowering the price to target sellers. Can you provide some intuition as well?

Hint: Warmup problem (i) supplies all the mathematical knowledge you will need to solve the remaining problems. What is needed for the rest is an ability to structure the problem, and an understanding of simple economic principles. The point of the exercise is to determine whether the boycott is effective in any of the three cases above. To answer this question, you have been given the following information: (1) the market supply curve, (2) the targeted fraction of sellers, (3) the market demand curve, (4) the observers as a fraction of buyers. Items (1) and (2) imply that it is possible to write down algebraic expressions for the supply curves of targeted and non-targeted sellers. You can also choose to use Excel to get a sense of how prices get determined in each sub-market.

Problem 2. Avocados

- a. In 1980, less than 1% of avocados in the US were imported. California supplied the vast majority of avocados grown in the US. The supply of avocados (in millions of pounds) from California is given by:

$$Q_S = \frac{1}{2}(250P - 100)$$

Demand for avocados (in millions of pounds) in the US is given by:

$$Q_D = 400 - 100P$$

If the US only has access to California avocados, what would the equilibrium price (for a pound of avocados) and quantity be?

Price: _____

Quantity: _____

- b. By 2019, avocado consumption had increased and most avocados consumed in the US were imported.¹ Mexico is by far the largest producer of avocados in the world. Given the favorable climate, Mexico is able to supply avocados at a price of \$1 per pound (and there is effectively no limit on how many can be grown).

If there is free trade with Mexico, how many millions of pounds of avocados will be consumed in the US, and what share of them will be produced in the US? Are US producers and consumers better or worse off from trade?

Quantity consumed: _____

US-produced Share: _____

- ☐ US consumers and US producers are better off
- ☐ US consumers better, US producers worse off
- ☐ US consumers worse, US producers better off
- ☐ US consumers and US producers worse off

¹ The true figure is approximately 90%.

- c. The government is considering imposing a tariff on imported avocados to help US farmers. Suppose they were to levy a tariff of \$0.20 per pound on imports. By how much would US consumer surplus increase or decrease from this tariff? By how much would US producer surplus increase or decrease from this tariff? How much government revenue would the tariff generate? (Hint: draw a diagram of the situation and compute areas corresponding to the different objects at the free-trade equilibrium and tariff equilibrium).

Change in US consumer surplus from tariff: _____

Change in US producer surplus from tariff: _____

Government revenue from tariff: _____

- d. Avocado lovers are not happy about the plan and decide to lobby against it. The government proposes to hand back 100% of the tariff revenue it collects to avocado consumers at the end of the year.² Would this make avocado lovers just as well-off as under free trade? Why?

☐ Yes

☐ No

☐ Need more information

Why? _____

² Ignore the practical challenges of doing this for now; assume they have a way to do this.